

AWS Networking Basics

Why use the cloud, and what its network looks like inside



Why the cloud

How it differs from on-premises

AWS infrastructure

Data centers, Availability Zones, Regions

How to draw it

Carve VPCs and subnets with CIDR.

01 On-premises

You buy the servers yourself

- You pay up front for all you bought, used or not.
- Run short, and you order more and wait for delivery.



On-premises



02 Cloud

You rent the servers

- Pay only for what you use.
- Start with one server; scale up and down as needed.



Cloud



What's different

—	On-premises	Cloud
Cost	Large up-front spend (CapEx)	Pay as you go (OpEx)
Capacity	Fixed at purchase	Change it when needed
Lead time	Order · ship · install	Minutes
Upkeep	Power, cooling, repairs — all yours	AWS runs buildings and hardware
Going global	Ship hardware into that country	Just pick a Region

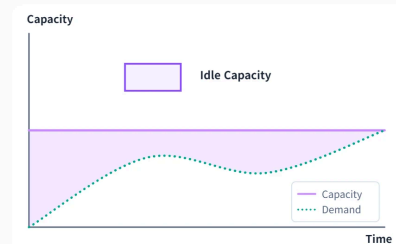
03 Idle servers

Idle Capacity

- On quiet days, the capacity you bought stays the same.
- You can't shrink capacity you've bought.
- Every unused bit is pure cost.



On-premises



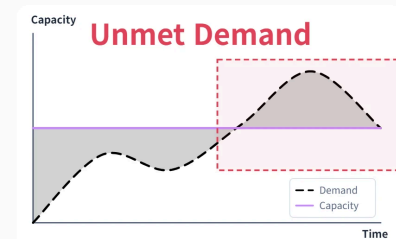
04 Peak days

Unmet Demand

- Demand can exceed the capacity you bought.
- Requests over that limit go unserved.
- Ordering more still means waiting for delivery and install.



On-premises



\$ Paid up front

before a single server boots

Equipment

Servers, storage and network gear, bought in advance

Facilities

Rack space, power, cooling and lines cost extra

People

Installs, replacements, 3 a.m. outages — your job

Time

The wait from running short until new gear arrives

🏢 When on-premises still makes sense

When regulation dictates location

Some industries must keep data inside specific buildings.

Specialized hardware

Some work needs gear the cloud doesn't rent out.

Hardware you already own

Servers not yet fully depreciated may be cheaper to keep.

05 Elasticity

Growing and shrinking to follow demand

- Elasticity means scaling up **and back down**.
- No more guessing capacity in advance.



✦ Six advantages of the cloud

[Exam point](#)

- ☐ Trade fixed expense for variable expense
- ☐ Stop guessing capacity
- ☐ Stop spending money running data centers
- ☐ Benefit from massive economies of scale
- ☐ Increase speed and agility
- ☐ Go global in minutes

Cloud isn't all-or-nothing. There are three deployment models: **cloud** runs everything in the cloud, **hybrid** works alongside existing hardware, and **on-premises** (private cloud) runs on your own gear only.

⚠ The cloud doesn't always cost less

Left running, it keeps billing

You pay for whatever is on, used or not.

Data transfer costs

Data coming in is mostly free; data going out is charged.

Moving as-is saves nothing

Moved as-is, idle servers stay idle. You need the right architecture and ops.

🛡 Who handles what

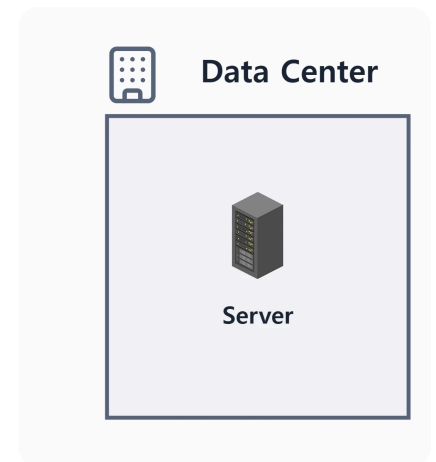
[Exam point](#)

—	AWS's part	Your part
Name	Security of the cloud	Security in the cloud
Protects	Facilities · hardware · Regions · AZs · virtualization	Guest OS, patches · apps · data
Network	Physical links and everything below	Security groups · routing · what's open

06 Data Center

One building packed with hundreds of servers

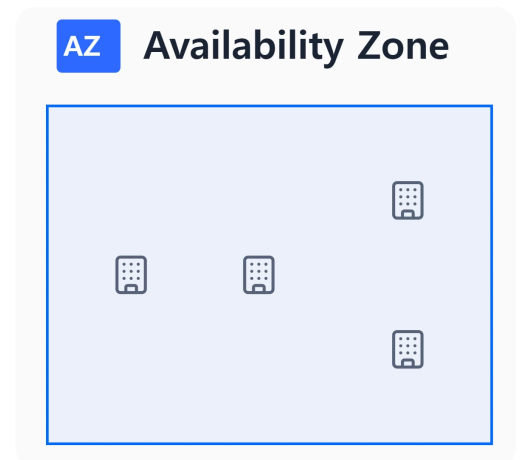
- The cloud is still physical servers racked somewhere.
- Each building has its own power, cooling and network.



07 Availability Zone

A group of nearby data centers

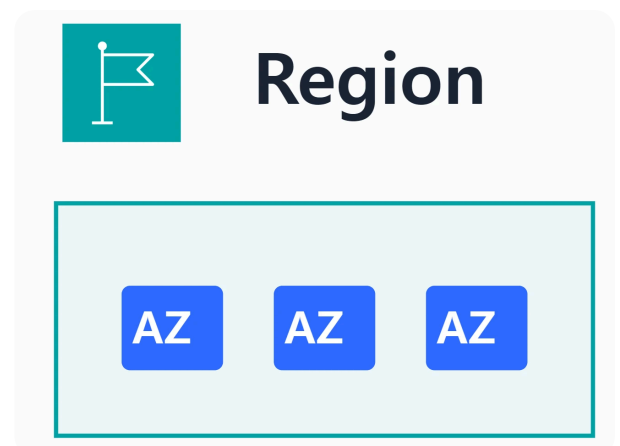
- Designed with disasters in mind
- Far enough apart that a power cut in one spares the rest (**within 100 km** of each other).
- Yet close enough that latency goes unnoticed.



08 Region

A group of Availability Zones

- Each Region has **three or more AZs**.
- 39 Regions · 124 AZs worldwide (2026).
- And still growing



How it stacks up

Server

One physical machine



Data center

One building



Availability Zone

AZ · within 100 km



Region

3+ AZs

All of this is built by AWS. What you build starts once you pick a Region.

Reading Region codes

area · direction · number

ap-northeast-1	Tokyo
ap-northeast-2	Seoul
ap-northeast-3	Osaka
us-east-1	N. Virginia

Area (ap · us · eu), then direction, then opening order in that area. Seoul is 2: Tokyo opened first.

Four factors in choosing a Region

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Close to your users

Distance is latency. Serving Korea? Pick Seoul.

Pricing

The same instance costs different amounts per Region.

Service availability

New services reach each Region at different times.

Compliance and data residency

Some data must stay in the country.

Other forms of AWS infrastructure

beyond Regions and AZs

Local Zones

A Region extended closer to users. 46 sites

Wavelength Zones

Inside carriers' 5G networks. 33 sites

Outposts

AWS hardware installed in your building

Edge locations

Where CloudFront caches. 750+ sites

⚠️ Region and AZ pitfalls

Regions are isolated

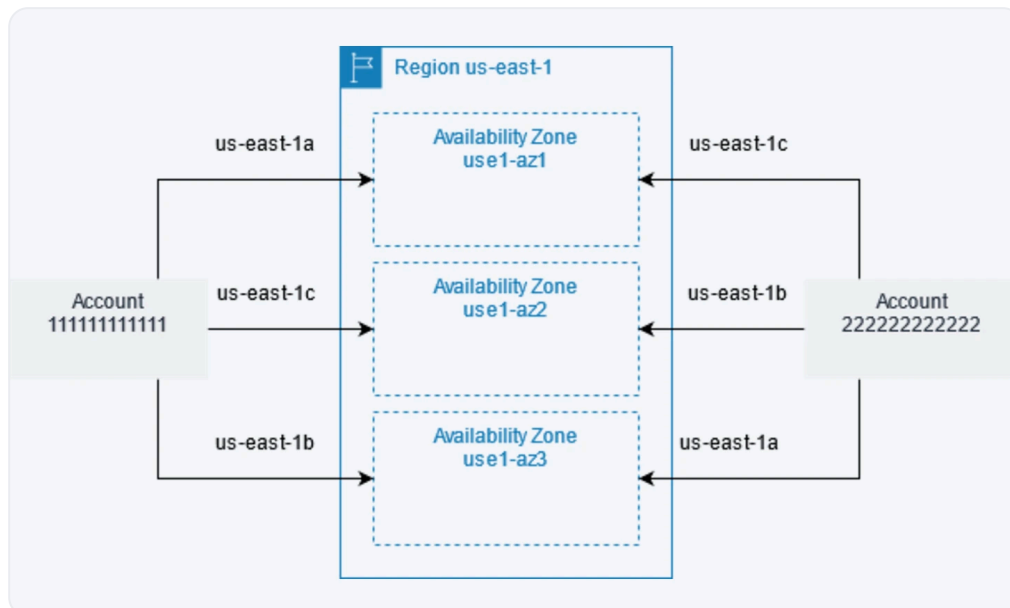
Check which Region you created a resource in.

a · b · c differ per account

The physical zone behind us-east-1a is shuffled randomly per account. To line up two accounts, use the **AZ ID** (use1-az1), not the name.

Cross-AZ traffic costs money

Even within a Region, \$0.01/GB is charged in each direction (free within one AZ).



Different accounts, different zones. The **AZ ID** points to the same place across accounts.

[AWS docs: "Availability Zone IDs"](#)

📦 Using AZs

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Spread resources over two or more

If one AZ goes down entirely, the service survives — the reason AZs exist.

One subnet per AZ

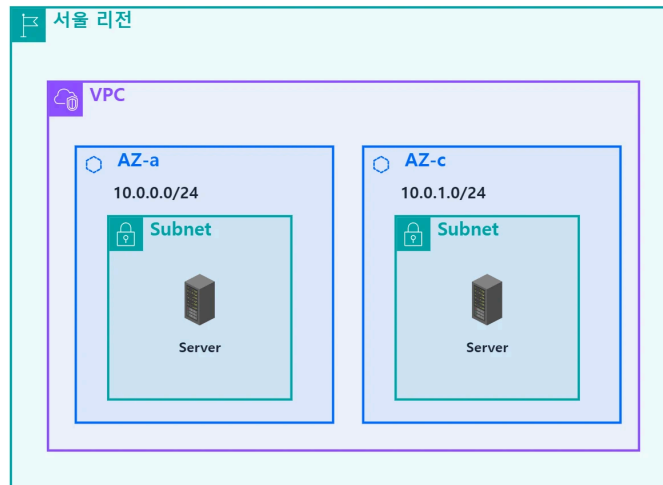
Redundancy starts with how you split subnets.

Across accounts

The console shows a·b·c. Use AZ IDs only to match zones with another account.

09 Working down

Pick a Region, create a VPC, split it into subnets



- A VPC is created in **one Region** and can't leave it.
- But it spans **all** AZs in that Region.
- A subnet belongs to **one AZ**.
- Sizes are assigned with CIDR.

What contains what

Region	3+ Availability Zones
VPC	One Region · all of its AZs

Availability Zone	1+ data centers
Subnet	One AZ · one VPC

Where services live

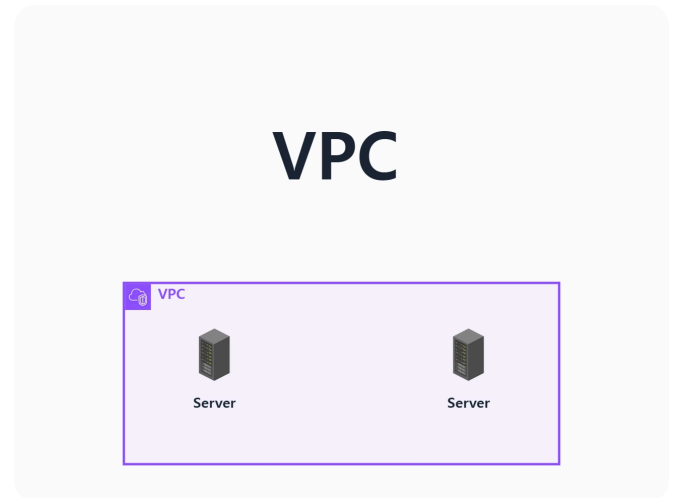
Exam point

Scope	Examples	What it means
Global	IAM · Route 53 · CloudFront	No Region to pick. Same from every Region
Regional	S3 · VPC · DynamoDB · Lambda	Spans all AZs in the Region
Zonal	EC2 instances · EBS volumes · subnets	Lives in one AZ. Goes down with it

10 VPC

Virtual Private Cloud

- A logically isolated virtual network in your account.
- No outside traffic gets in until you open a door.



VPC rules

One Region

It stays in the Region you create it in (but spans all its AZs).

Pick addresses from private ranges

RFC 1918 — 10.0.0.0/8 · 172.16.0.0/12 · 192.168.0.0/16. AWS recommends these.

A default VPC already exists

Every account has one per Region, so you can launch instances right away.

5 per Region

Default quota. Request an increase to go up to hundreds.

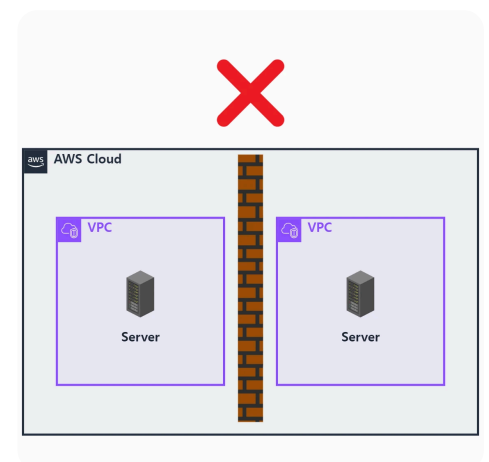
The VPC itself is free

You pay for services like NAT gateways, and for public IPv4 addresses.

11 VPC to VPC

Blocked by default

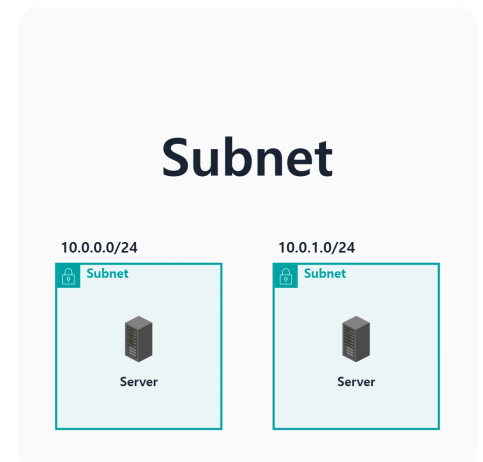
- VPCs are fully isolated; servers in another VPC are out of reach.
- To connect them, set up VPC peering or a Transit Gateway.



12 Subnet

A slice of the VPC's IP range

- Carve a 10.0.0.0/16 VPC into pieces like 10.0.0.0/24.
- Every server sits inside some subnet.



Subnet rules

[Exam point](#)

One AZ

A subnet belongs to exactly one AZ. It can't span two.

Redundancy starts here

The first step is one subnet in each AZ.

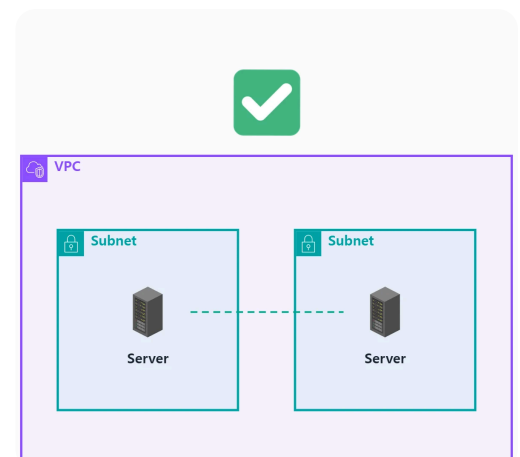
200 per VPC

Default quota. Can be raised on request.

13 Subnet to subnet

Allowed within the same VPC

- A new VPC's route table gets a local route for its whole range.
- To block it, use security groups or network ACLs.



Public vs. private subnets

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Public subnet	Its route table has a route to an internet gateway
Private subnet	No route to an internet gateway. Outbound traffic goes through NAT
What decides it	Not a creation option — just whether its route table has an internet gateway route!

14 CIDR

Classless Inter-Domain Routing

- Notation for taking only what you need, no classes (1993).
- Classful addressing forced blocks far bigger than needed.

CIDR

10.0.0.0 / 16

192.168.0.0 / 24

15 Octet

8 bits, so 256 values

- 8 bits can express $2^8 = 256$ numbers.
- An IPv4 address is **four** octets
- Four dot-separated numbers — four octets
- 192.168.0.0

Octet

⁸
= 2

= 256

16 After the slash

Prefix — how many leading bits are fixed

- The fixed front is the **network**; the rest are **hosts** in it.
- With /24, the first 24 bits are fixed and the last 8 give 192.168.0.0 ~ 192.168.0.255.

192.168.0.0 / 24

11000000.10101000.00000000.00000000

11000000.10101000.00000000.11111111

Private IP ranges

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Block	Range	Example
10.0.0.0/8	10.0.0.0 ~ 10.255.255.255	10.0.0.0/16
172.16.0.0/12	172.16.0.0 ~ 172.31.255.255	172.31.0.0/16
192.168.0.0/16	192.168.0.0 ~ 192.168.255.255	192.168.0.0/20

 VPCs and subnets: /16 to /28**Exam point**

Prefix	Addresses	Usable IPs
/16	65,536	65,531
/20	4,096	4,091
/24	256	251
/28	16	11

Reserved addresses in a subnet

first four and the last one

10.0.0.0	Network address
10.0.0.1	VPC router
10.0.0.2	DNS
10.0.0.3	Reserved (future use)
10.0.0.255	Broadcast

 Before choosing a CIDR**Must fall inside the VPC range**

A 10.0.0.0/16 VPC can't hold a 192.168.0.0/24 subnet.

Subnets can't overlap

Each address belongs to one subnet. Overlapping CIDRs are rejected.

Go big from the start

Spare addresses cost nothing. /24 is a safe subnet size.

 CIDR pitfalls**A CIDR can't be resized once set**

To grow, add a secondary CIDR (5 per VPC); to shrink, you must create a new VPC.

Nothing smaller than /28

Only 11 of 16 addresses are usable — already tight in practice.

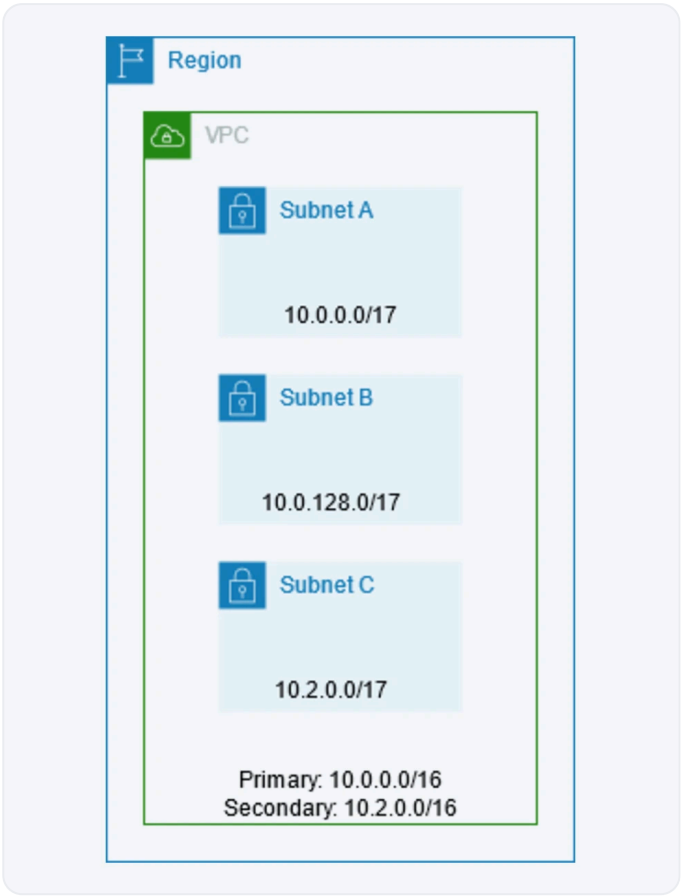
⌚ Ranges to avoid

Rejected outright

0.0.0.0/8 · 127.0.0.0/8 · 169.254.0.0/16 · 224.0.0.0/4 can't be used for a VPC.

Avoid 172.17.0.0/16

Services like Cloud9 and SageMaker use it. Overlap it and you can't connect to them.



A VPC with a **secondary CIDR** added. Only subnet C uses the new range — added, not resized.

[AWS docs: "VPC CIDR blocks"](#)

🌐 How IPv6 differs

you can't pick the addresses

VPC	One /56 from Amazon. You can't choose which	Subnet	/64 — not split finely like IPv4
Reserved	Five, like IPv4 — first four, last one	Private ranges	None in IPv6. Routing decides what goes out

 Try it!

Practice

1 A regulation says data must stay in the country. What is the first step to comply on AWS?

- ☐ A Availability Zone
- ☐ B Region
- ☐ C Subnet
- ☐ D VPC

2 A `10.0.0.0/16` VPC in the Seoul Region gets one subnet in each of two AZs. Which subnet can't be made?

- ☐ A `10.0.0.0/24` — ap-northeast-2a
- ☐ B `10.0.1.0/24` — ap-northeast-2c
- ☐ C `192.168.0.0/24` — ap-northeast-2a
- ☐ D `10.0.2.0/28` — ap-northeast-2c

3 Which two statements about Regions and AZs are true? 2 answers

- ☐ A A subnet can span multiple AZs
- ☐ B A VPC exists within a single Region
- ☐ C us-east-1a is the same physical zone in every account
- ☐ D Regions are isolated, so resources aren't replicated automatically
- ☐ E An AZ is just another name for one data center

4 At most how many EC2 instances fit in one `10.0.0.0/24` subnet?

- ☐ A 256
- ☐ B 255
- ☐ C 254
- ☐ D 251

Answers and explanations

- 1 B The Region decides which country the data is in. An AZ is a location *inside* the chosen Region, so it can't change the country; VPCs and subnets are drawn after the Region is chosen.
- 2 C A subnet CIDR must sit inside the VPC CIDR. `192.168.0.0/24` is outside `10.0.0.0/16`, so it's rejected. The other three are in range and don't overlap, and `/28` is the smallest size allowed.
- 3 B · D A subnet belongs to one AZ only, and the names a·b·c map randomly to physical zones per account (the AZ ID is what stays fixed). An AZ groups **one or more** data centers.
- 4 D A `/24` has 256 addresses, but AWS takes five in every subnet: the first four (network · VPC router · DNS · reserved) and the last (broadcast). The familiar 254 forgets three of those first four.

Numbers to remember

Exam point

3**Min AZs per Region**

Usually more

100km**Distance between AZs**

Kept no farther apart

/16**Largest CIDR**

65,536 addresses

/28**Smallest CIDR**

16 addresses

5**Reserved per subnet**

First four and the last one

251**Usable IPs in a /24**

256 - 5

5**VPCs per Region**

Default quota, adjustable

200**Subnets per VPC**

Default quota, adjustable

 Glossary

every term on this sheet, one line each

Region A geographic group of 3+ AZs**AZ ID** Same physical zone in any account (use1-az1)**Subnet** A slice of a VPC. Lives in exactly one AZ**CIDR** Notation giving size after a slash**Octet** 8 bits. An IPv4 address is four octets**Elasticity** Scaling up with demand, then back down**Availability Zone (AZ)** 1+ data centers, ≤ 100 km apart**VPC** An isolated virtual network you create in a Region**Public subnet** Has a route to an internet gateway**Prefix** The number after the slash: fixed leading bits**Private IP** The 3 RFC 1918 ranges; not internet-routed**On-premises** Buying servers and running them yourself Up next

building paths inside a VPC

Route tables

Where traffic goes; decides public vs. private

Internet gateway

The VPC's door to the internet

NAT gateway

Outbound-only exit for private subnets

Security groups · NACLs

Firewalls in front of instances and of subnets

In one line

Data center → AZ → Region are the physical layers AWS builds; on top, you carve VPC → subnets with CIDR to draw your own network.

Data Center

AZ

Region

VPC

Subnet

CIDR

Official sources

[AWS docs](#) · [standards](#) · [videos](#) · [articles](#)

-  **AWS Global Infrastructure — Regions and AZs** [↗](#) [pp. 4 · 6](#)
Source of 39 Regions · 124 AZs, 3+ AZs, and the 100 km figure
aws.amazon.com/about-aws/global-infrastructure/regions_az
-  **Six advantages of cloud computing** [↗](#) [p. 3](#)
The six names, exactly as AWS words them
docs.aws.amazon.com/whitepapers/latest/aws-overview/six-advantages-of-cloud-computing.html
-  **Shared Responsibility Model** [↗](#) [p. 3](#)
The line between security of and in the cloud
aws.amazon.com/compliance/shared-responsibility-model
-  **Regions and Zones (EC2 User Guide)** [↗](#) [pp. 6 · 7](#)
Region isolation, AZ codes, Local Zones and Outposts
docs.aws.amazon.com/AWSEC2/latest/UserGuide/using-regions-availability-zones.html
-  **Availability Zone IDs** [↗](#) [p. 6](#)
Why a · b · c are shuffled per account, and use1-az1
docs.aws.amazon.com/ram/latest/userguide/working-with-az-ids.html
-  **What is Amazon VPC?** [↗](#) [pp. 8 · 9](#)
Defines VPCs, subnets and routing — and says the VPC itself is free
docs.aws.amazon.com/vpc/latest/userguide/what-is-amazon-vpc.html
-  **Default VPCs** [↗](#) [p. 8](#)
What the VPC already in your account comes with
docs.aws.amazon.com/vpc/latest/userguide/default-vpc.html
-  **VPC CIDR blocks** [↗](#) [pp. 11 · 12](#)
/16 ~ /28, RFC 1918 recommended; no resizing, but a secondary CIDR can be added
docs.aws.amazon.com/vpc/latest/userguide/vpc-cidr-blocks.html
-  **Subnet CIDR blocks** [↗](#) [p. 11](#)
The five addresses reserved in each subnet, one by one
docs.aws.amazon.com/vpc/latest/userguide/subnet-sizing.html
-  **Amazon VPC quotas** [↗](#) [pp. 8 · 9](#)
5 VPCs per Region; 200 subnets and 5 CIDRs per VPC
docs.aws.amazon.com/vpc/latest/userguide/amazon-vpc-limits.html
-  **RFC 1918 — private IP ranges** [↗](#) [p. 12](#)
Where 10 · 172.16 · 192.168 come from. A standard, not an AWS doc
datatracker.ietf.org/doc/html/rfc1918
-  **Overview of Data Transfer Costs for Common Architectures** [↗](#) [p. 6](#)
Diagrams of what is charged across AZs and Regions
aws.amazon.com/blogs/architecture/overview-of-data-transfer-costs-for-common-architectures
-  **Networking basics every AWS user must know** [↗](#) [pp. 4 · 8](#)
Same scope as this sheet, in Korean. Official AWS Korea channel
www.youtube.com/watch?v=vCNexbgYmQ8
-  **AWS Networking Fundamentals (re:Invent 2025)** [↗](#) [p. 14](#)
Official one-hour session on the next steps (in English)
www.youtube.com/watch?v=nXBqxnP5ybY